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**Minor Project Report
on**

“Blockchain Based Certificate Verification System”

22MIT461P

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(Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi)

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CERTIFICATE

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DECLARATION

I, **Harshith B**, student of third semester M.Tech in **Information Technology, Department of ISE**, RV College of Engineering®, Bengaluru declare that the Minor Project titled “**Blockchain Based Certificate Verification System**”, has been carried out by me. It has been submitted in partial fulfilment for the award of degree in **Master of Technology** in Information Technology of RV College of Engineering®, Bengaluru, affiliated to Visvesvaraya Technological University, Belagavi, during the academic year **2023-24**. The matter embodied in this report has not been submitted to any other university or institution for the award of any other degree or diploma.

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ABSTRACT

The proliferation of digital certificates and credentials has given rise to concerns surrounding authenticity, fraud, and the complexities of manual verification processes. Traditional centralized systems are vulnerable to single points of failure, data tampering, and lack of transparency. This application proposes a blockchain-based solution to address these challenges by leveraging the immutable, decentralized, and transparent nature of distributed ledger technology.

The proposed system comprises three main components: a user-friendly interface for certificate submission, integration with a blockchain network for secure data storage and hash generation, and a web page dedicated to certificate verification. Users can submit their digital certificates and relevant details through an intuitive interface, which generates a unique cryptographic hash value representing the certificate's contents.

This hash value is recorded on a blockchain network, ensuring its immutability and resistance to tampering. To facilitate efficient verification, the hash value is encoded into a QR code associated with the certificate. During the verification process, users can scan or upload the QR code, which is decoded to extract the hash value. The system queries the blockchain to retrieve the recorded hash and compares it with the extracted value. If the hash values match, the certificate is deemed authentic and valid. However, if a mismatch occurs, it indicates potential tampering, and the certificate is flagged as invalid.

The system adheres to security, performance, and usability best practices, ensuring a seamless user experience. It is designed to be scalable, interoperable, and promotes broader adoption across educational institutions and professional organizations globally. By leveraging blockchain's inherent advantages, this system establishes a trusted, transparent, and tamper-evident framework for digital certificate verification, reducing fraud and enhancing the credibility of academic and professional credentials.

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Chapter 1

INTRODUCTION

Blockchain technology has emerged as a revolutionary force, disrupting various industries and offering unprecedented levels of transparency, security, and decentralization. In the realm of digital certificates, blockchain-based verification systems have gained significant traction, addressing the long-standing challenges of fraud, counterfeiting, and authenticity concerns. The traditional methods of issuing and verifying certificates have long been plagued by vulnerabilities, such as centralized databases susceptible to hacking, complex verification processes, and the potential for human error. These limitations have given rise to the need for a more secure, efficient, and trustworthy system, which blockchain technology can provide. A blockchain-based certificate verification system leverages the inherent strengths of distributed ledger technology, ensuring the immutability and transparency of recorded data. By integrating digital certificates with blockchain networks, each certificate is assigned a unique hash value, which serves as a digital fingerprint, capturing the essence of the certificate's content and metadata. This hash value is then recorded on the blockchain, making it virtually impossible to alter or tamper with without leaving a traceable trail. The decentralized nature of the blockchain eliminates the need for a central authority, thereby reducing the risk of a single point of failure and increasing the overall security and resilience of the system. Furthermore, the integration of QR codes into the verification process enhances the accessibility and convenience of certificate verification. Users can simply scan the QR code associated with a certificate, which encodes the corresponding hash value, to initiate the verification process seamlessly. By leveraging the immutable and transparent nature of blockchain technology, digital certificates can be verified with a high degree of confidence, reducing the risk of fraud and counterfeiting. The integration of QR codes and user-friendly interfaces simplifies the verification process, making it accessible to a wide range of users, from educational institutions to professional organizations. The decentralized architecture of blockchain networks eliminates the need for a central authority, enhancing security and reducing the risk of a single point of failure. Any attempt to modify or alter a recorded certificate on the blockchain will result in a mismatch between the hash values, providing an immediate and tamper-evident record. As the demand for secure and reliable digital credentials continues to grow, blockchain-based certificate verification

systems emerge as a compelling solution, offering a robust framework for establishing trust, transparency, and authenticity in the digital realm.

Overview

Blockchain-based certificate verification offers a secure and transparent solution for validating digital certificates' authenticity. By integrating blockchain technology, each certificate is assigned a unique hash value recorded on an immutable distributed ledger. This hash value acts as a digital fingerprint, ensuring data integrity. The system converts these hash values into QR codes for convenient verification. Users can scan QR codes to initiate the process, where the system retrieves the certificate data from the blockchain and compares the generated hash with the recorded one. If they match, the certificate is authenticated; if not, it's flagged as potentially tampered. This approach leverages blockchain's strengths, enhancing trust, efficiency, and tamper-evidence while streamlining verification through user-friendly interfaces and QR code integration.

Literature Review

Blockchain technology has a wide range of applications, such as education, engineering, administration, medicines, elections, construction, and e-government. In this research, we primarily focus on the education field. Therefore, we present some research cases of the using of block chain technology. 1 EDSA The European Data Science Academy (EDSA) is a good example of the application of blockchain technology for providing data science skills to job seekers. EDSA also helps provide training to a new generation of world leading data scientists. In the 21st century, data is being produced from all industries at a phenomenal rate, which has introduced numerous challenges regarding the collection, storage, and analysis of the data.

[1] D. A. Gayathiri, J. Jayachitra (2022). Certificate validation using blockchain. 7th International Conference on Smart Structures and Systems ICSSS 2020, 8, 34929 – 34941. This paper proposes a blockchain-based system for certificate validation to address issues like certificate forgery and manual verification challenges. It highlights the immutable and decentralized nature of blockchains that can ensure transparency and tamper-resistance.

[2] A. S. P. A. Fouzia F. Ozair, Nayer Jamshed (2021). "Award badging and validation method using blockchain. This paper focuses on using blockchains for validating digital badges or micro-

credentials awarded for skills, achievements or certifications. It argues that blockchain transparency can overcome challenges in traditional badging like centralization and lack of trusted verification.

[3]Min choi and Shinde Rupali Kiran and Se-Chang oh and oh-young and Kwon. First, generate the electronic file of a paper certificate accompanying other related data into the database, meanwhile calculate the electronic file for its hash value. Finally, store the hash value into the block in the chain system. The system will create a related QR-code and inquiry string code to affix to the paper certificate. It will provide the demand unit to verify the authenticity of the paper certificate through mobile phone scanning or website inquiries. Through the unmodifiable properties of the blockchain, the system not only enhances the credibility but also loss risks of various types of certificates.

[4] Hongning Dai Xiangping Chen Zibin Zheng Shaoan Xie¹ and Huaimin Wang. Blockchain is a sequence of blocks, which holds a complete list of transaction records like conventional public ledger . Figure 1 illustrates an example of a blockchain. With a previous block hash contained in the block header, a block has only one parent block. It is worth noting that uncle blocks (children of the block's ancestors) hashes would also be stored in Ethereum blockchain . The first block of a blockchain is called genesis block which has no parent block. We then explain the internals of blockchain in details Each user owns a pair of private key and public key. The private key that shall be kept in confidentiality is used to sign the transactions. The digital signed transactions are broadcasted throughout the whole network. The typical digital signature is involved with two phases: signing phase and verification phase.

[5] A Gayatri and J Jayachitra and Drs matilda, Sitalakshmi Venkataraman In today's digital world, everything is digitalized in which the certificate of SSLC, HSC, and academic certificate are digitalized in educational institution and provided to the students. Students are facing difficulties to maintain their degree certificates. For the organization and institution, validation of certificates are tedious and cumbersome. Our project will help to store the certificate in the blockchain system and provide security. First, the paper certificates are converted into digital certificates. The chaotic algorithm is used to generate the hash code value for the certificate. Then the certificates are stored in blockchain generate the digital certificate for the paper certificate then hash value created for the certificate is stored in the blockchain system.

“Microsoft exam and certification badge” Microsoft partnered with Pearson VUE’s Acclaim platform to provide a badge for certification acquisition. The badge is a Microsoft certification mark that is available on the web and consists of images and unique metadata associated with the certificate holder. The badge provides the owner detailed information about the technology and demonstrates possession of the technology. The benefits of Microsoft badges are as follows; easily share certification and test results with an expert network identify who employs the technology holder 3. identify the expected salary of the workplace that requires the Microsoft skills possessed by the badge holder, and search for job-related announcements related to certification details and complete support with just a few clicks.

Motivation

The motivation behind implementing a blockchain-based certificate verification system stems from the growing need to address the challenges associated with traditional methods of issuing and verifying digital certificates. These challenges include the susceptibility of centralized databases to hacking, complex verification processes, and the potential for human error. Additionally, the rise in cases of fraud and counterfeiting of certificates has highlighted the importance of establishing a secure and transparent system that can ensure the authenticity and integrity of digital credentials. By leveraging the immutable and decentralized nature of blockchain technology, this system aims to restore trust, increase efficiency, and provide a tamper-evident solution for validating digital certificates.

1.1 Problem Statement

The traditional methods of issuing and verifying digital certificates have been plagued by numerous vulnerabilities and inefficiencies. Centralized databases housing certificate records are susceptible to hacking, data manipulation, and single points of failure. Manual verification processes are often complex, time-consuming, and prone to human error. Moreover, the increasing prevalence of fraudulent activities, such as counterfeiting and altering certificates, has eroded trust in digital credentials. These challenges underscore the need for a robust and secure system that can ensure the authenticity and integrity of digital certificates throughout their lifecycle. A blockchain-based verification system addresses these issues by leveraging the immutable, decentralized, and transparent nature of distributed ledger technology, providing a tamper-evident and efficient

solution for validating digital certificates, thereby restoring trust and confidence in the certification process.

1.2 Objectives

To develop a user-friendly interface allowing individuals/organizations to submit certificates and relevant details. To develop integration with a blockchain network for generating unique cryptographic hash values per certificate. To develop a mechanism for recording these hash values on the immutable blockchain network.

To implement a QR code generation component that encodes the recorded hash values into machine-readable QR codes for efficient verification. To implement a dedicated web page for certificate verification, enabling users to upload or scan associated QR codes. To implement a decoding process that extracts the hash value from the submitted QR code.

To implement a querying mechanism that retrieves the recorded hash value from the blockchain based on certificate details. To implement a comparison function that validates the certificate's authenticity by matching the extracted and recorded hash values. To implement a user-friendly display of the verification result, indicating if the certificate is authentic or potentially tampered with. To ensure adherence to security, performance, and usability best practices throughout development and deployment.

Scope

The scope of the blockchain-based certificate verification system encompasses the entire lifecycle of digital certificates, from issuance to verification. It covers the development of a user-friendly interface for submitting certificates, the integration with a blockchain network for secure data storage and hash value generation, the conversion of hash values into QR codes for convenient verification, and the creation of a dedicated web page for authenticating certificates. The system is designed to be scalable and interoperable, capable of handling a large volume of certificate submissions and verifications from various educational institutions, professional organizations, and regulatory bodies. It aims to establish a universal standard for digital certificate verification, promoting trust, transparency, and accountability across diverse sectors and geographical regions.

1.3 Methodology

Task 1: User Interface Development:

- Design and develop a user-friendly web interface or application for users to submit their digital certificates and relevant details.
- Implement form fields for users to enter personal information such as name, institution, degree, and year of graduation.
- Incorporate a file upload functionality to allow users to submit their certificate files in various formats.
- Implement input validation and error handling mechanisms to ensure data integrity and provide a smooth user experience.

Task 2: Blockchain Integration and Hash Value Generation

- Establish a connection with a blockchain network (e.g., Ethereum, Hyperledger Fabric) using appropriate libraries or frameworks.
- Develop a function to generate a cryptographic hash value (e.g., SHA-256) from the certificate data and metadata provided by the user.
- Record the generated hash value on the blockchain network, leveraging its immutable and decentralized nature to ensure data integrity and transparency.

Task 3: QR Code Generation

- Integrate a QR code generation library (e.g., qrcode) into the system. - Implement a function to convert the recorded hash value into a QR code image.
- Associate the generated QR code image with the corresponding certificate for future verification purposes.

Task 4: Certificate Verification Web Page

- Develop a dedicated web page or section within the application for certificate verification.
- Implement functionality to capture the QR code image, either through file upload or direct scanning using a device's camera.
- Decode the QR code image to extract the encoded hash value.

- Query the blockchain network to retrieve the recorded hash value associated with the certificate details.
- Compare the extracted hash value from the QR code with the recorded hash value on the blockchain.
- Display the verification result to the user, indicating whether the certificate is authentic (if the hash values match) or potentially tampered with (if the hash values do not match).
- Implement appropriate error handling and user feedback mechanisms for a smooth verification experience.

Task 5: Deployment, Testing, and Maintenance

- Deploy the blockchain-based certificate verification system to a suitable hosting environment (e.g., cloud platform, local server).
- Conduct thorough testing of the user interface, blockchain integration, and verification processes, including edge cases and error scenarios.
- Ensure adherence to best practices for security, performance, and usability. - Implement mechanisms for monitoring, logging, and maintenance to ensure the system's ongoing operation and support.
- Develop strategies for scalability and capacity management to accommodate potential increases in user base and certificate volume.
- Explore potential integrations with other systems or platforms (e.g., educational institutions, professional organizations) to promote broader adoption and interoperability.

1.4 Organization of the Project

The project report is organized into nine chapters, each addressing specific aspects of the Blockchain based certificate verification System. Chapter 1 provides an introduction to the project, highlighting the significance of automated assessment systems in educational settings and outlining the objectives and scope of the project. Chapter 2 presents the System Requirements Specification (SRS), detailing the functional and non-functional requirements of the assessment system. Chapter 3 focuses on the High-Level Design, discussing the architectural design, system components, and data flow within the system. Chapter 4 delves into the Implementation Details, providing insights into the algorithms, libraries, and technologies used to develop the assessment

system. Chapter 5 covers the Testing and Validation procedures, including unit testing, integration testing, and user acceptance testing. Chapter 6 presents the Evaluation and Results, analyzing the performance and effectiveness of the assessment system based on predefined criteria. Chapter 7 discusses the Deployment Strategies, outlining the process of deploying the system in educational settings and ensuring its ongoing maintenance and support. Chapter 8 offers a comprehensive Discussion and Analysis, synthesizing the findings from the evaluation and reflecting on the implications for future research and development. Finally, Chapter 9 concludes the report with a Summary and Conclusion, summarizing the key findings, contributions, and implications of the project, as well as suggesting potential avenues for further exploration.

Chapter 2

THEORY AND CONCEPTS OF BLOCKCHAIN BASED VERIFICATION SYSTEM

2.1 Concepts Related to the Domain of Work

1. Blockchain: Blockchain is a decentralized, distributed digital ledger that records transactions across numerous computers in a secure and immutable manner using cryptography, enabling transparent and tamper-proof record-keeping without relying on a central authority.

2. Graphical User Interface (GUI): A GUI is a type of user interface that allows users to interact with electronic devices through graphical icons and visual indicators. PyQt5 is a popular framework for building GUI applications in Python.

3. Hash Verification: Hash verification is the process of generating a hash value from data and comparing it to a previously recorded hash value to ensure the data's integrity and authenticity. If the hashes match, the data is considered unmodified and verified as authentic.

2.2 System Architecture/Framework/Models

1. Model-View-Controller (MVC) Architecture: The project follows the MVC architecture, where the data (Model), user interface (View), and application logic (Controller) are separated into distinct components. This architecture promotes code organization, modularity, and maintainability.

2. PyQt5 Framework: PyQt5 is a set of Python bindings for the Qt application framework, which is widely used for developing cross-platform desktop applications. It provides a comprehensive set of tools and widgets for creating interactive GUIs.

3. HashLibBcPython: HashLibBcPython is a Python library for building blockchain applications, providing functionalities for creating and managing blockchain networks, generating and validating blocks, and implementing consensus algorithms. Web3.py is another widely-used

Python library that enables interaction with Ethereum-based blockchain networks, facilitating the development of decentralized applications (DApps) and smart contracts.

4. QRcode: This is a pure Python library that allows you to generate QR codes with a simple interface. It supports several error correction levels and can encode various data types like text, URLs, and binary data.

2.3 Applications

1. Certificate Issuance: - The educational institution or certifying authority collects relevant details from the recipient (e.g., name, degree, year of graduation). - These details are combined and hashed using a cryptographic function like SHA-256 to generate a unique hash value. - This hash value is recorded on a blockchain network, ensuring immutability and transparency. - The `qrcode` library is used to generate a QR code encoding the hash value. - The QR code is printed on the physical certificate or provided as a digital file along with the certificate.

2. Certificate Verification: - A dedicated web application is developed for certificate verification. - The recipient or verifier scans the QR code on the certificate using the web application. - The web application decodes the QR code to extract the hash value. - The application queries the blockchain network to retrieve the recorded hash value for the given certificate details. - If the extracted hash value matches the recorded hash value on the blockchain, the certificate is considered authentic and valid. - If the hash values do not match, it indicates potential tampering or fraud, and the certificate is flagged as invalid.

3. Data Integrity and Auditability: - Since the hash value is recorded on the blockchain, any attempt to modify the certificate details will result in a different hash value. - This mismatch between the extracted and recorded hash values immediately alerts the system to potential tampering. - The blockchain provides an immutable and transparent audit trail, allowing stakeholders to verify the authenticity of certificates at any given time.

4. Interoperability and Global Recognition: - The blockchain-based verification system can be adopted by multiple educational institutions, professional bodies, and regulatory agencies globally. By adhering to a common standard and leveraging the decentralized nature of blockchain, certificates issued by different entities can be seamlessly verified and recognized across borders. -

This promotes trust, transparency, and efficiency in the certification process, benefiting both certificate holders and verifiers. The integration of blockchain technology, cryptographic hashing, QR codes, and user-friendly web applications enables a secure, tamper-evident, and globally accessible solution for verifying the authenticity of digital certificates, reducing fraud and enhancing the credibility of educational and professional credentials.

2.4 Summary

The blockchain-based certificate verification system generates immutable hash values from certificate data and records them on a distributed ledger. QR codes encoding these hashes are associated with certificates. During verification, the QR code is scanned, the hash extracted, and compared against the blockchain record to validate authenticity, ensuring trust and preventing fraud.

Chapter 3

SOFTWARE REQUIREMENT SPECIFICATION

This chapter outlines the software requirements for the project, including the overall description and specific requirements.

3.1 Overall Description

3.1.1 Product Perspective

The software is a standalone desktop application designed to provide users with functionalities related to Blockchain Hashing, qr code scanner, and interaction. It operates independently on users' local machines without requiring an internet connection.

3.1.2 Product Functions

- 1. Certificate Issuance and Registration:** - Provide a secure and tamper-proof method for educational institutions or certification bodies to issue and register certificates on the blockchain.
- 2. Certificate Verification:** - Enable employers, verification authorities, or any interested parties to easily verify the authenticity and validity of certificates. Implement a user-friendly interface where verifiers can input the certificate details or scan a QR code to initiate the verification process.
- 3. Certificate Management and Revocation:** - Provide authorized issuers with the ability to manage and update certificate records on the blockchain, if needed.

3.1.3 Constraints

Operating System Compatibility: The application is compatible with Windows, macOS, and Linux operating systems.

Dependencies: Requires Python 3.9 and the installation of additional libraries such as PyQt5, pandas, blockchain, qrcode.

Resource Requirements: The application may require sufficient system resources, especially for large datasets and complex computations.

3.2 Specific Requirements

3.2.1 Product Functions

- 1. Certificate Issuance and Registration:** - Provide a secure and tamper-proof method for educational institutions or certification bodies to issue and register certificates on the blockchain.
- 2. Certificate Verification:** - Enable employers, verification authorities, or any interested parties to easily verify the authenticity and validity of certificates. Implement a user-friendly interface where verifiers can input the certificate details or scan a QR code to initiate the verification process.
- 3. Certificate Management and Revocation:** - Provide authorized issuers with the ability to manage and update certificate records on the blockchain, if needed.

3.2.2 Performance Requirements

Responsiveness: The application should respond to user inputs promptly, with minimal latency.

Scalability: Should be able to handle datasets of varying sizes efficiently without significant performance degradation.

3.2.3 Software Requirements

Python 3.9: The application is written in Python and requires Python 3.9 to run.

PyQt5: Dependency for GUI development.

Blockchain: Creates blocks for the given data and generates Hash Value.

Qrcode: Generates QR code for the Hash Value.

3.2.4 Hardware Requirements

Processor: Intel core i3

Memory: At least 4GB of RAM recommended for optimal performance.

Storage: Sufficient disk space to store the application and datasets.

3.2.5 Design Constraints

User Interface Design: The application's GUI should be intuitive and user-friendly, catering to users with varying levels of technical expertise.

Code Modularity: The codebase is modular and well-organized, facilitating ease of maintenance and future enhancements.

3.3 Summary

In summary, the software requirement specification outlines the overall description of the project, including its perspective, functions, and constraints. It also details specific requirements related to functionality, performance, software, hardware, and design constraints. By adhering to these requirements, the project aims to deliver a robust and user-friendly desktop application for text processing and natural language understanding.

Chapter 4

HIGH LEVEL DESIGN

In this chapter, An overview of the high-level design considerations, architecture strategies, and data flow diagrams for the project.

4.1 Design Considerations

4.1.1 General Considerations

User-Focused Design: Prioritize user experience and usability in the design of the graphical user interface (GUI).

Modularity: The application in a modular fashion to promote code reusability, maintainability, and scalability.

Compatibility: Compatibility with multiple operating systems (Windows, macOS, Linux) to reach a broader user base.

4.1.2 Development Methods

Agile Development: Adopt an iterative and incremental approach to development, allowing for flexibility and adaptation to changing requirements.

Test-Driven Development (TDD): Emphasize writing tests before implementing functionality to ensure code correctness and improve software quality.

4.2 Architecture Strategies

4.2.1 System Architecture

Model-View-Controller (MVC) Architecture: Divide the application into three interconnected components: Model (data), View (user interface), and Controller (application logic). This architecture promotes separation of concerns and facilitates code organization and maintenance as shown in Fig 4.1

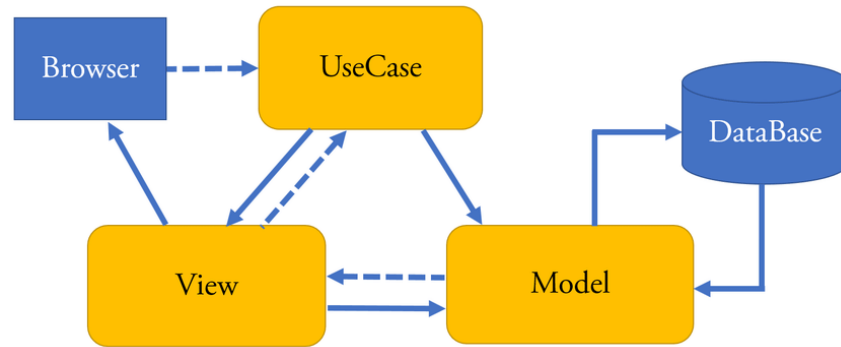


Fig 4.1 MVC Model

4.2.2 Architectural Style

Layered Architecture: Implement a layered architectural style where different components of the application are organized into horizontal layers. This allows for better encapsulation, abstraction, and separation of concerns.

4.3 Data Flow Diagram

Fig 4.2 shows the dataflow for the project

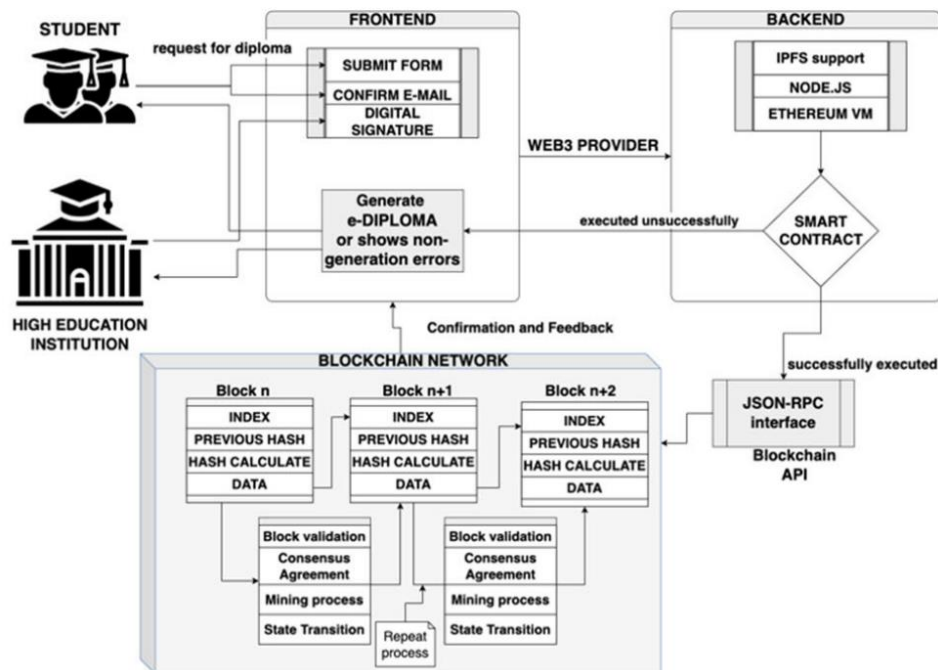


Fig 4.2 Dataflow diagram

The process of generating a certificate in electronic form and the process of verification of the certificate in a higher education institution. The blockchain system for the generation, verification, and revocation of certificate would include the following actors: students, higher education institutions, the blockchain network, and the application part of the system, which is Front End and Back End. Communication between students and institutions for higher education is done through the application part, namely Front End and Back End. Front End represents the design part of the system, making it easy to use both by the student and the academic staff of the higher education institution. The Back End is the part in the background that processes all the requests that are made in the Front-End part, where it converts all the requests and directs them to be executed. Whereas a blockchain network represents the database, and the platform where all transactions are executed. Students after having fulfilled all their obligations to the institution of higher education, i.e., have finished their studies, they have the option to request the generation of a diploma. All this is done through the application part of the blockchain system. Which is most important during the student's request is the attachment of the student's digital signature, in addition to other data that accurately identifies the student. The essence of the operation of this system is the checking of hash values. Every service that has been executed before in the blockchain system has certain hash values that are stored in the blockchain network. Therefore, in order to generate certificate, a necessary control of the services and their values is necessary, including all the courses attended, assessments made, payments, etc. The respective institution can only generate the diploma in electronic form and digitally sign it after confirming that the student who made the request has all of the unique hash values created at the beginning. The same process is also used for verification of the certificate.

4.4 Summary

In summary, the high-level design of the project emphasizes user-focused design, modularity, and compatibility. The system architecture follows the MVC pattern, while the architectural style is based on a layered approach. Data flow diagrams provide a visual representation of the system's data flow and interactions at different levels of detail. These design considerations and strategies lay the foundation for the successful implementation of the project, ensuring robustness, maintainability, and scalability.

Chapter 5

DETAILED DESIGN

5.1 Structure Chart

The structure chart illustrates the hierarchical structure of the application, showing the relationships between different modules and submodules. It provides a visual representation of the system's architecture and organization, helping to understand the flow of control and data within the application as shown in fig 5.1

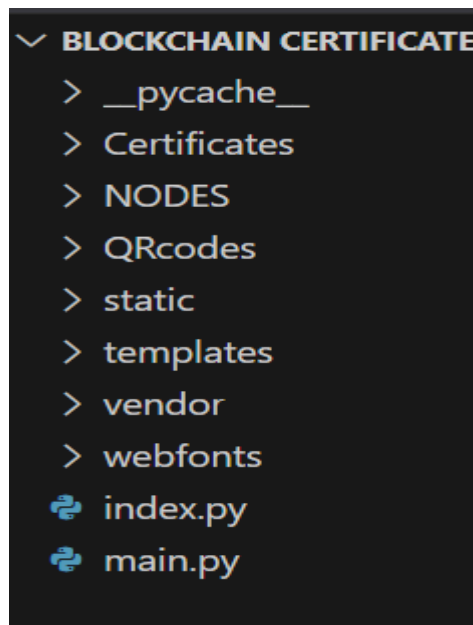


Fig 5.1 Structure chart

5.2 Functional Description of the Modules

5.2.1 Certificates (Folder)

Description: This module handles and stores the certificates and the information of the student.

Functionality: Accepts and stores the user information in Hash formate

5.2.2 Nodes (app2.py file)

Description: This module holds the ethereum block nodes.

Functionality: Retrieves nodes and create hash for the given information.

5.3 Summary

In summary, the detailed design of the project includes a structure chart illustrating the hierarchical structure of the application and the functional description of individual modules. These modules collectively provide the core functionalities of the application, catering to diverse user needs and use cases.

Chapter 6

IMPLEMENTATION

6.1 Programming Language Selection

Python

Reasoning: Python was chosen due to its simplicity, readability, and versatility. It offers a wide range of libraries and frameworks for GUI development (PyQt5 in this case), data manipulation (Pandas), and natural language processing (SentenceTransformers).

Benefits: Python's syntax is straightforward, making it easy to understand and learn. It has a large and active community, extensive documentation, and a rich ecosystem of third-party libraries.

Relevance: Python is widely used in various domains including web development, data science, machine learning, and automation, making it a suitable choice for developing applications with diverse requirements.

6.2 Platform Selection

Desktop Application (Windows, VS Code)

Reasoning: A desktop application provides a native user interface and can run directly on users' computers without the need for a web browser. This ensures a consistent user experience across different operating systems (Windows).

Benefits: Desktop applications offer offline access, better performance, and greater control over system resources compared to web applications. They can also integrate seamlessly with the underlying operating system and hardware.

Target Audience: The application targets users who prefer standalone software solutions with a familiar desktop environment.

6.3 Code Conventions

6.3.1 Naming Conventions

Variables: Follow `lower\_case\_with\_underscores` convention for variable names.

Functions and Methods: Use `lower\_case\_with\_underscores` convention for function and method names.

Classes: Use `UpperCamelCase` convention for class names.

6.3.2 File Organization

Module Structure: Each major functionality or component should be organized into separate modules (Python files).

Directory Structure: Maintain a clear directory structure with related files grouped together. For example, separate directories for GUI components, data processing, and utilities.

6.3.3 Properties Declarations

Explicit Declaration: Properties should be explicitly declared and documented, specifying their data type and purpose.

Visibility: Follow appropriate visibility modifiers (public, protected, private) depending on the property's intended usage.

6.3.4 Class Declarations

Class Structure: Class definitions should follow a clear structure with attributes, methods, and constructors organized logically.

Documentation: Include descriptive docstrings for classes, methods, and properties to explain their purpose, parameters, and return values.

By adhering to these code conventions, the codebase will be more readable, maintainable, and consistent, facilitating collaboration and future development effort as shown in fig 6.1

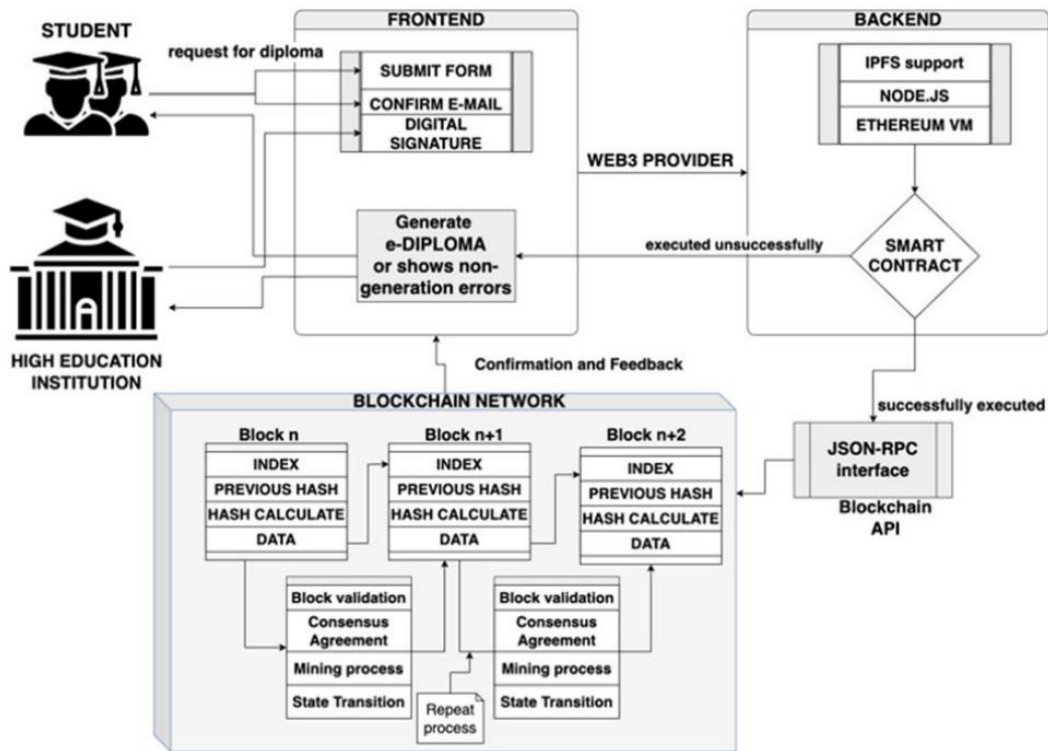


Fig 6.1 Implementation

Fig 6.1, we have adapted a simplified architecture of a blockchain system, which represents the process of generating a certificate in electronic form. The same could be adapted for the process of verification of the certificate in a higher education institution. The blockchain system for the generation, verification, and revocation of certificate would include the following actors: students, higher education institutions, the blockchain network, and the application part of the system, which is Front End and Back End. Communication between students and institutions for higher education is done through the application part, namely Front End and Back End. Front End represents the design part of the system, making it easy to use both by the student and the academic staff of the higher education institution. The Back End is the part in the background that processes all the requests that are made in the Front-End part, where it converts all the requests and directs them to be executed. Whereas a blockchain network represents the database, and the platform where all transactions are executed. Students after having fulfilled all their obligations to the institution of higher education, i.e., have finished their studies, they have the option to request the generation of a diploma. All this is done through the application part of the blockchain system. Which is most

important during the student's request is the attachment of the student's digital signature, in addition to other data that accurately identifies the student. The essence of the operation of this system is the checking of hash values. Every service that has been executed before in the blockchain system has certain hash values that are stored in the blockchain network. Therefore, in order to generate certificate, a necessary control of the services and their values is necessary, including all the courses attended, assessments made, payments, etc. The respective institution can only generate the diploma in electronic form and digitally sign it after confirming that the student who made the request has all of the unique hash values created at the beginning. The same process is also used for verification of the certificate.

Chapter 7

SOFTWARE TESTING

7.1 Test Environment

Operating Systems: Windows 10

Python Version: Python 3.9, VS Code IDE

Dependencies: PyQt5, pandas, sentence_transformers, pytest (for unit testing)

7.2 Unit Testing of Main Modules

Modules for Unit Testing:

1. `SimpleUI`: GUI components and event handling methods.
2. `Hash\_value`: Hashing functionality.
3. `qrcode`: qr functionality.

Unit Testing Approach:

Use the pytest framework for writing and executing unit tests.

Write test cases to cover various scenarios for each main module.

Verify the expected behavior of individual functions and methods.

7.3 Integration Testing of Modules

Modules for Integration Testing:

- `SimpleUI` with `hash\_value`, `qrcode`.

Integration Testing Approach:

Test the interaction between different modules to ensure they work together seamlessly.

Verify that the GUI components interact correctly with the backend functionalities.

Check for any data flow issues or communication errors between modules.

7.4 System Testing

Conduct end-to-end testing of the entire application.

Verify the application's behavior in different environments and on different operating systems.

Test all features and functionalities to ensure they meet the specified requirements. Check for any performance issues, compatibility issues, or security vulnerabilities.

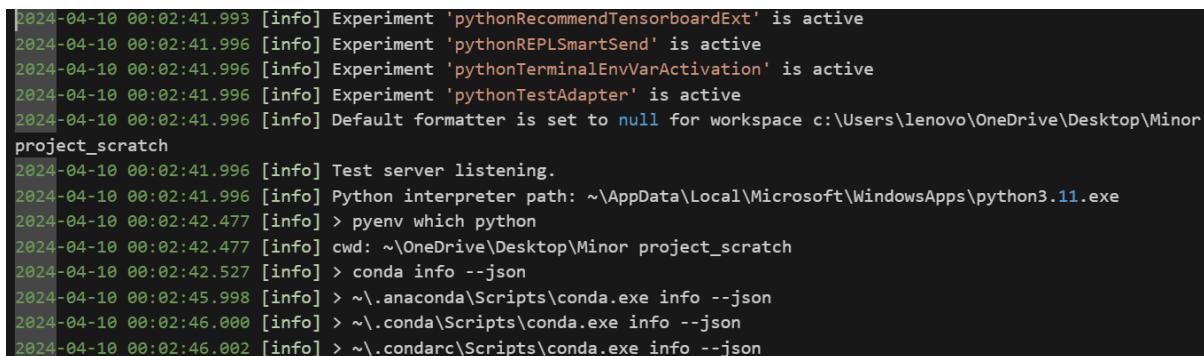
7.5 Functional Testing

Test each function and method to ensure it performs its intended task correctly.

Verify that user interactions produce the expected outcomes.

Check for error handling and edge cases to ensure robustness.

Test all input fields, buttons, and other GUI components for proper functionality.



```
2024-04-10 00:02:41.993 [info] Experiment 'pythonRecommendTensorboardExt' is active
2024-04-10 00:02:41.996 [info] Experiment 'pythonREPLSmartSend' is active
2024-04-10 00:02:41.996 [info] Experiment 'pythonTerminalEnvVarActivation' is active
2024-04-10 00:02:41.996 [info] Experiment 'pythonTestAdapter' is active
2024-04-10 00:02:41.996 [info] Default formatter is set to null for workspace c:\Users\lenovo\OneDrive\Desktop\Minor
project_scratch
2024-04-10 00:02:41.996 [info] Test server listening.
2024-04-10 00:02:41.996 [info] Python interpreter path: ~\AppData\Local\Microsoft\WindowsApps\python3.11.exe
2024-04-10 00:02:42.477 [info] > pyenv which python
2024-04-10 00:02:42.477 [info] cwd: ~\OneDrive\Desktop\Minor project_scratch
2024-04-10 00:02:42.527 [info] > conda info --json
2024-04-10 00:02:45.998 [info] > ~\.anaconda\Scripts\conda.exe info --json
2024-04-10 00:02:46.000 [info] > ~\.conda\Scripts\conda.exe info --json
2024-04-10 00:02:46.002 [info] > ~\.condarc\Scripts\conda.exe info --json
```

Fig 7.1 Testing

7.6 Summary

All main modules were thoroughly tested using unit testing techniques as shown in Fig 7.1. Integration testing was conducted to ensure the seamless interaction between different modules. System testing was performed to validate the application's behavior in various environments.

Chapter 8

EXPERIMENTAL ANALYSIS AND RESULTS

To provide an experimental analysis and results of the project, we'll discuss the result analysis, experimental datasets, evaluation metrics, performance analysis, inference of the results obtained, and summarize the findings.

8.1 Result Analysis

Analyzing the outcomes of various functionalities. Evaluate the accuracy and effectiveness of each functionality based on user interactions and system responses. Identify any issues or discrepancies encountered during testing and usage. Fig 8.1 shows the login page of the application.

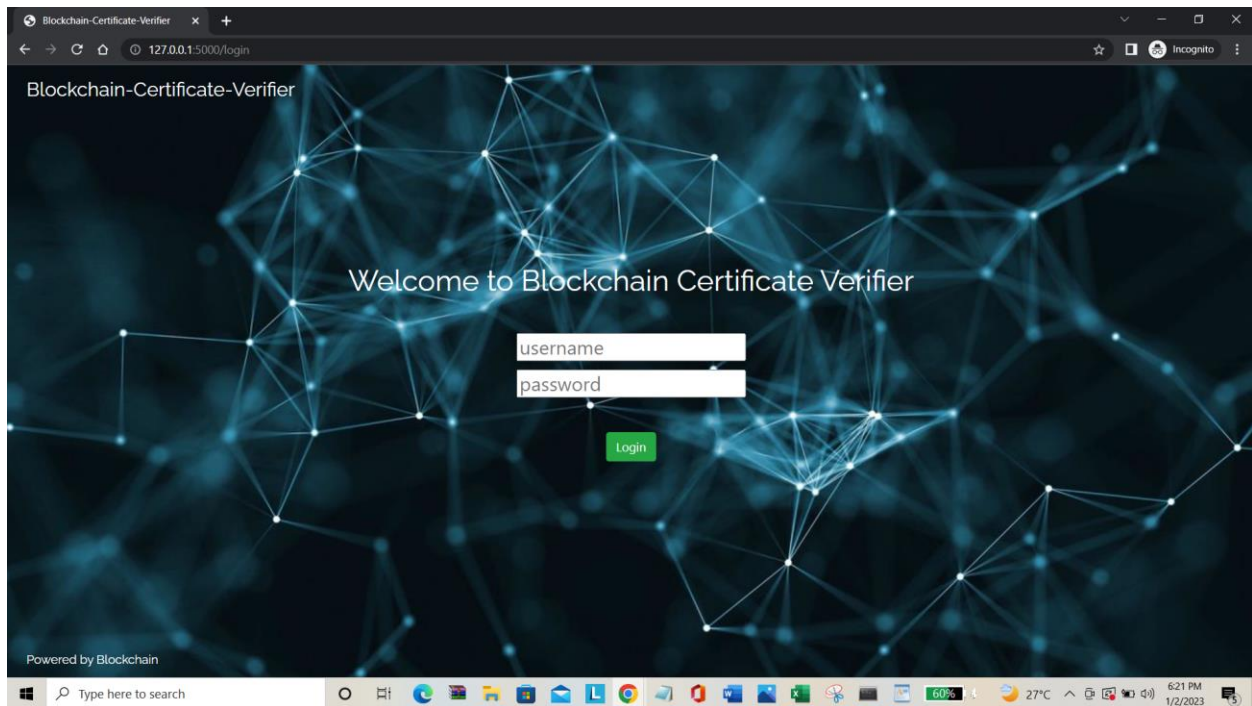


Fig 8.1 GUI Application

The user must enter correct username and password in order to login.

To Add certificate, the user can click and add the certificate and the information

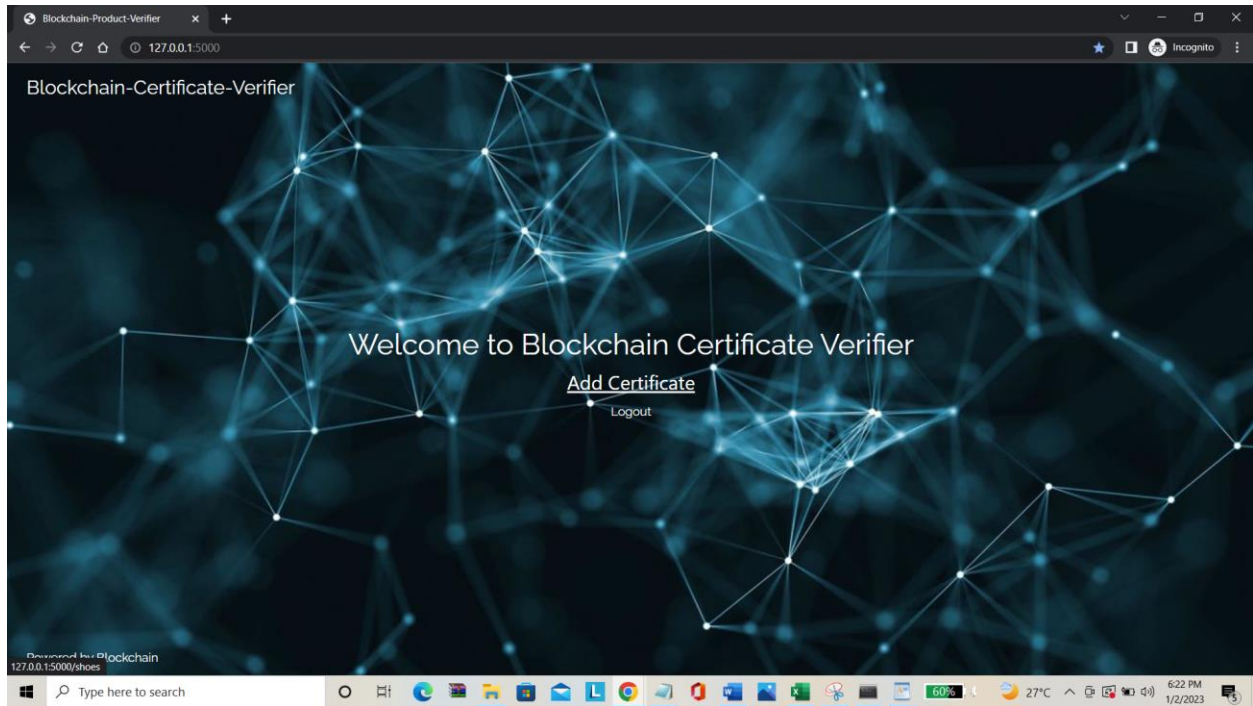


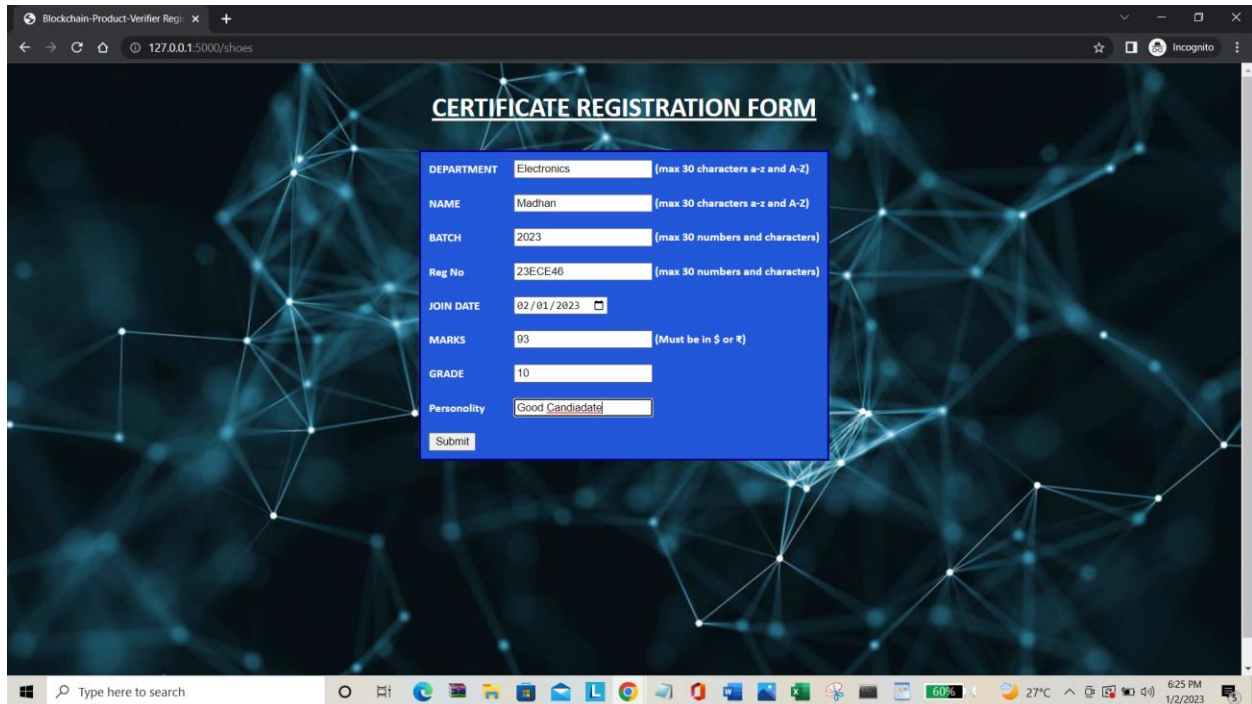
Fig 8.2 Blockchain Verify

Fig 8.3 shows certificate registration form

A screenshot of the 'CERTIFICATE REGISTRATION FORM' within the same web application. The form is a white box with a blue border, containing several input fields. The fields are labeled: 'DEPARTMENT', 'NAME', 'BATCH', 'Reg No', 'JOIN DATE', 'MARKS', 'GRADE', and 'Personality'. Each field has a placeholder text indicating the maximum length or format (e.g., '(max 30 characters a-z and A-Z)', '(max 30 numbers and characters)', '(Must be in S or R)'). A 'Submit' button is located at the bottom of the form. The background of the page remains the same dark blue network graph pattern.

Fig 8.3 Certificate Registration Form

Certificate registration form will contain all the information of the student and the certificate.



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000/shoes". The page title is "Blockchain-Product-Verifier Regi". The main content is a "CERTIFICATE REGISTRATION FORM" with a blue background and a network diagram. The form fields are as follows:

Field	Value	Constraints
DEPARTMENT	Electronics	(max 30 characters a-z and A-Z)
NAME	Madhan	(max 30 characters a-z and A-Z)
BATCH	2023	(max 30 numbers and characters)
Reg No	23ECE46	(max 30 numbers and characters)
JOIN DATE	02/01/2023	
MARKS	93	(Must be in \$ or %)
GRADE	10	
Personality	Good Candidate	

A "Submit" button is located at the bottom of the form.

Fig 8.4 Click submit

The information is stored in hash of the blockchain output as shown in fig 8.5

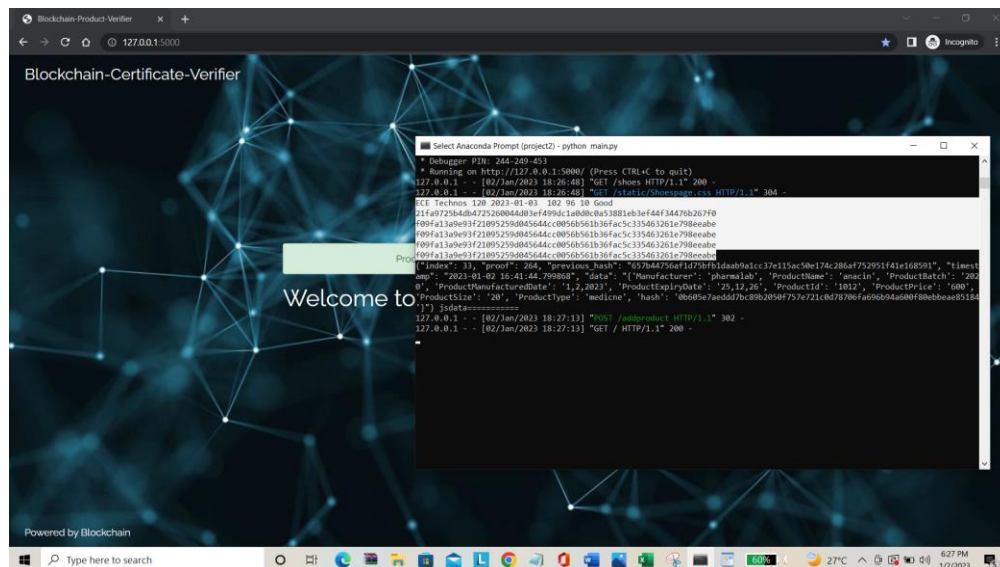


Fig 8.5 Block Created

This is the QR code that is generated through Hash value

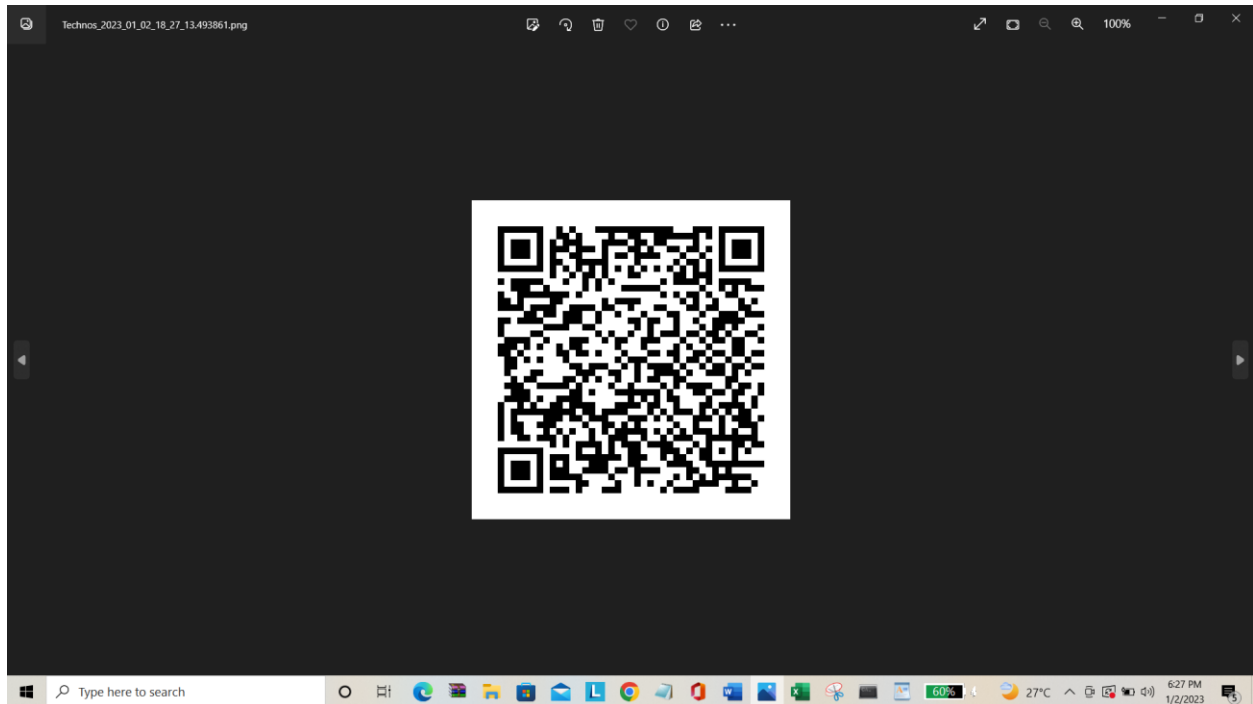


Fig 8.6 Generated QR code

To verify, click the link in QR scanner and this link will redirect to other Tab.

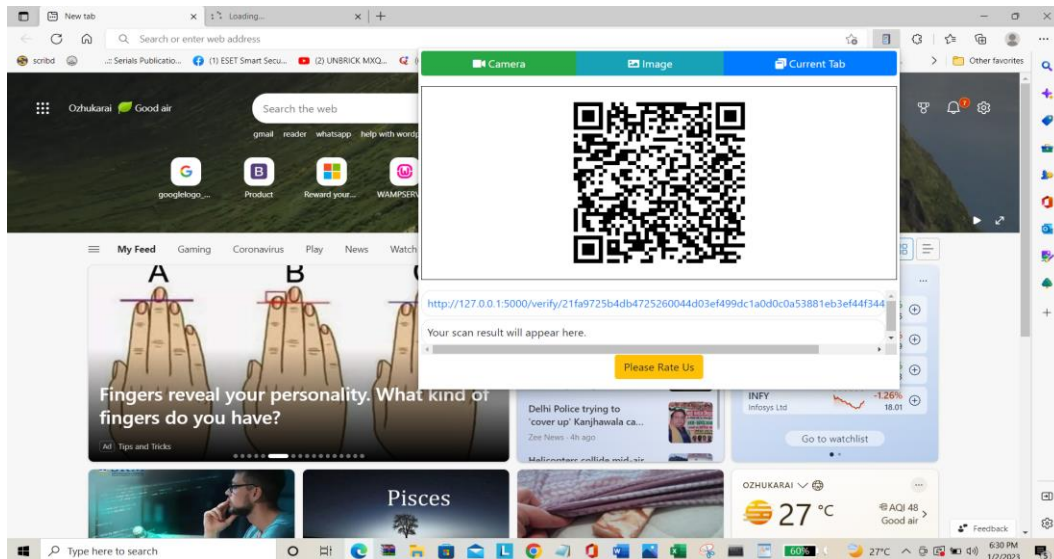


Fig 8.7 QR code scanner

This page will verify the has which is been scanned by the QR code Scanner

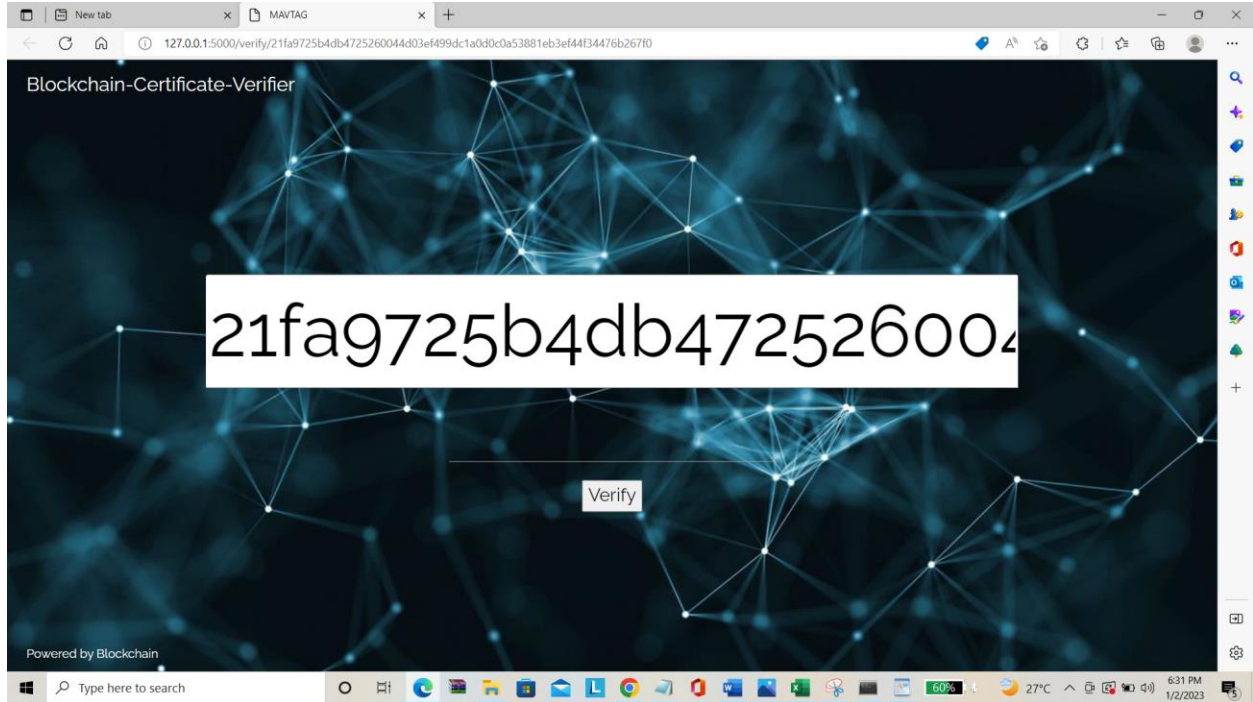


Fig 8.8 Block Verifier

If, the certificate is authentic, the blockchain will verify and give you the result

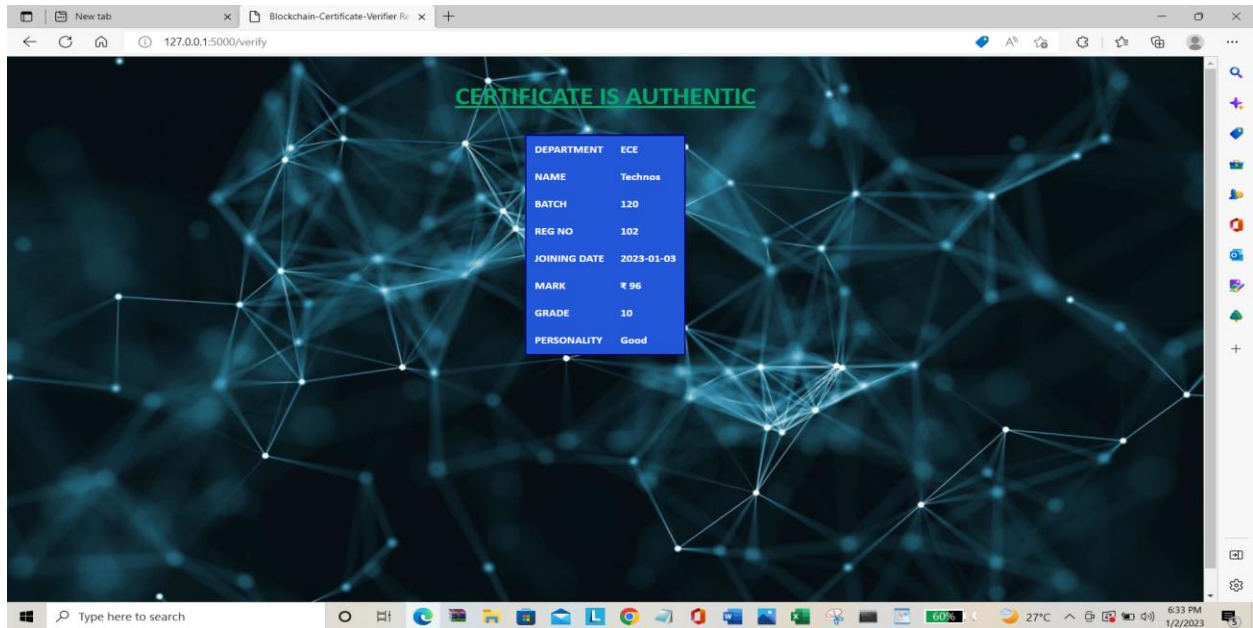


Fig 8.9 Certificate is Authentic

8.2 Experimental Datasets and Evaluation Metric

Experimental Hash Node:

- Use sample datasets for testing the functionalities. These datasets should contain a variety of input values and questions to cover different scenarios.
- For similarity calculation, use a set of sentence pairs with known similarity scores for evaluation purposes.

Evaluation Metric:

- For the search functionality, evaluate the precision, recall, and F1 score based on the retrieved results compared to the ground truth.
- For the quiz functionality, calculate the accuracy of user responses compared to the correct answers.
- For similarity calculation, use evaluation metrics such as cosine similarity score and Pearson correlation coefficient to measure the similarity between sentence pairs.

8.3 Performance Analysis

- Measure the application's performance in terms of responsiveness, resource usage, and execution time.
- Evaluate the memory usage and CPU utilization during different operations such as search, quiz, and similarity calculation.
- Identify any bottlenecks or performance issues that may affect the user experience.

8.4 Inference of the Results Obtained

Inference of Results:

- Interpret the analysis and evaluation results to draw conclusions about the effectiveness and efficiency of the application.
- Discuss any limitations or constraints encountered during testing and usage.

8.5 Summary

- Summarize the experimental analysis and results obtained from testing the project.
- Highlight key findings, including the performance metrics, evaluation outcomes, and inference drawn from the results.
- Discuss the implications of the findings and their significance in the context of the project objectives.
- Conclude with recommendations for future enhancements or modifications based on the observed results and insights gained.

By conducting a comprehensive experimental analysis and results evaluation, we can gain valuable insights into the project's performance, effectiveness, and areas for improvement, ultimately enhancing the overall quality and usability of the software application.

Chapter 9

CONCLUSION

In conclusion, while blockchain-based certificate verification systems offer transparency, immutability, and decentralization, they face challenges like scalability, privacy concerns, and adoption barriers. Future enhancements focus on improving scalability, implementing privacy-preserving techniques, establishing interoperability standards, adopting sustainable consensus mechanisms, enhancing user experience, integrating with emerging technologies, and ensuring regulatory compliance. Addressing these limitations and capitalizing on potential enhancements will drive the widespread adoption and effectiveness of these systems.

Through ongoing development, collaboration, and feedback, we aim to create a robust and user-friendly software solution that delivers value and facilitates learning and exploration in various domains.

9.1 Limitations

1. Scalability: Blockchain technology has scalability challenges, particularly when dealing with a large volume of transactions or data. As the number of certificates and verifications increases, the network may face performance issues and slower transaction processing times.

2. Privacy and Data Protection: While blockchain provides transparency, there are concerns about the privacy of personal and sensitive information stored on a public ledger. Appropriate measures must be taken to protect individuals' privacy and comply with data protection regulations.

3. Adoption and Interoperability: Widespread adoption and interoperability among different organizations and industries can be challenging. Standardization and common protocols are necessary to ensure seamless integration and acceptance of blockchain-based certificate verification systems.

4. Energy Consumption: Some blockchain consensus mechanisms, like Proof-of-Work (PoW), are energy-intensive and may not align with sustainability goals. Alternative consensus mechanisms, such as Proof-of-Stake (PoS), can help mitigate this issue.

5. User Experience: Blockchain technology can be complex for end-users, requiring additional training and education to ensure proper usage and understanding of the certificate verification process.

9.2 Future Enhancements

1. Improved Scalability: Ongoing research and development in blockchain technology aim to address scalability concerns through solutions like sharding, off-chain computations, and layer-2 scaling solutions.

2. Privacy-Preserving Techniques: Advancements in cryptographic techniques, such as zero-knowledge proofs and secure multi-party computation, can enhance privacy while maintaining transparency and verification capabilities.

3. Interoperability Standards: The development of industry-wide standards and protocols for blockchain-based certificate verification systems can facilitate seamless integration and cross-organizational collaboration.

4. Sustainable Consensus Mechanisms: Continued research and adoption of energy-efficient consensus mechanisms, like Proof-of-Stake (PoS) or novel mechanisms, can contribute to the sustainability and environmental friendliness of blockchain solutions.

5. User Experience Improvements: Simplified user interfaces, educational resources, and user-friendly tools can enhance the accessibility and adoption of blockchain-based certificate verification systems for various stakeholders.

6. Integration with Emerging Technologies: Exploring the integration of blockchain with other emerging technologies, such as the Internet of Things (IoT), artificial intelligence (AI), and digital identities, can unlock new possibilities and use cases for certificate verification and authentication.

7. Regulatory Compliance: Ongoing collaboration with regulatory bodies and policymakers can ensure that blockchain-based certificate verification systems comply with relevant laws and regulations, facilitating broader adoption and trust.

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